

PACKET DECODER LAB

Decoding Traffic: A Hands-On Guide to Transport-Layer Protocols

Ten practical exercises for breaking down UDP, TCP, and QUIC packets at the byte level.

SET 1 Dissecting UDP Packets

Three UDP packets examined field by field.

Exercise 1 — DNS Request

Host 192.168.56.11 issues a DNS lookup for ns2.highsecure.ru, directed at 194.190.124.17.

The screenshot displays the Hex Packet Decoder web application. The URL bar shows <https://hpd.gasmi.net/?sid=DNS>. The interface includes buttons for 'Decode', 'Upload', and 'Clear', along with a 'Samples' dropdown. The packet description is '192.168.56.11 → 194.190.124.17 DNS Standard query 0xe4a3 A ns2.highsecure.ru OPT'. A button 'Ask AI about this packet' is also present. The main display is a hex dump of the packet data, organized into 16 columns (0-15). The data is color-coded by protocol layer: Ethernet (orange), IPv4 (blue), UDP (red), and DNS (green). Below the hex dump, it states '4 protocols in packet (88 bytes):' and lists 'Ethernet', 'IPv4', 'UDP', and 'DNS'. At the bottom, there is a 'Filter tree...' section showing a list of protocols: 'Frame 1: 88 bytes on wire (704 bits)', 'Ethernet II', 'Internet Protocol Version 4', 'User Datagram Protocol', and 'Domain Name System (query)'. The Windows taskbar at the bottom shows the date and time as 14:40 on 19-08-2026.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
52	54	00	7E	A2	A5	52	54	00	A6	11	53	00	00	45	00
00	45	0A	83	00	00	3C	11	F9	07	C0	A0	38	00	C2	0E
7C	11	E7	3E	00	35	00	36	37	CB	E4	A3	00	10	00	01
00	00	00	00	00	01	01	0C	73	32	0A	62	62	67	00	73
65	65	75	72	65	02	72	75	00	00	01	00	01	00	00	29
10	00	00	00	00	00	00	00								

4 protocols in packet (88 bytes):
Ethernet IPv4 UDP DNS

Filter tree... Width: 16 bytes Export Share

- Frame 1: 88 bytes on wire (704 bits)
- Ethernet II
- Internet Protocol Version 4
- User Datagram Protocol
- Domain Name System (query)

Exercise 2 — NTP Response

216.27.185.42 sends a time-sync reply back to 192.168.50.50. Notice both endpoints use port 123.

The screenshot shows the Hex Packet Decoder web application interface. At the top, the URL is `https://hpd.gasmi.net/?sid=NTP`. The application title is "Hex Packet Decoder" with a subtext "12,100,304 packets decoded". Below the title, there is a text area containing the raw hex data of the packet:

```
00 00 59 6C 40 4E 00 0C 41 82 B2 53 00 00 45 00 00 4C A2 22 40 00 32 11 22 5E 08 18 B9 3A C0 A8 32 32 00 7B 00 7B 00 30 70 E1 1A 02 0A EE 00 00 07 A4 00 00 0B A3 A4 43 3E C2 C5 02 01 B1 E5 79 18 19 C5 02 04 EC EC 42 EE 92 C5 02 04 EB D9 5E 8D 54 C5 02 04 EB D9 69 B1 74
```

Below the hex data, there are buttons for "Decode", "Upload", "Clear", and a "Samples" dropdown. The decoded packet information is displayed below:

216.27.185.42 → 192.168.50.50 NTP Version 3, symmetric passive

Below this, there is a table representing the packet structure. The table has 16 columns (0 to 15) and 4 rows (0 to 3). The data is as follows:

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	D0	59	6C	40	4E	00	0C	41	82	B2	53	00	00	45	00	00
01	00	5C	02	22	A0	00	32	11	22	5E	08	18	B9	3A	C0	A8
02	32	32	00	7B	00	7B	00	38	70	E1	1A	02	0A	EE	00	00
03	07	A4	00	00	A3	A4	43	3E	C2	C5	02	01	B1	E5	79	18
04	18	19	C5	02	04	EC	EC	42	EE	92	C5	02	04	EB	D9	5E
05	8D	54	C5	02	04	EB	D9	69	B1	74						

Below the table, there is a "4 protocols" section with a dropdown menu showing "User Datagram Protocol", "Src Port: 123", "Dst Port: 123", and "Length: 8 bytes". The protocols listed are Ethernet, IPv4, UDP, and NTP.

At the bottom, there is a "Filter tree..." input, a "Width: 16 bytes" dropdown, and "Export" and "Share" buttons.

Exercise 3 — NTP Response, UDP Header Detail

The same packet as above, now with the UDP header opened up to reveal the length field.

Hex Packet Decoder 12,100,304 packets decoded

00 D0 59 6C 40 4E 00 0C 41 B2 B2 53 00 00 45 00 00 4C A2 22 40 00 32 11 22 5E 08 18 B9 3A C0 A8 32 32 00 7B 00 7B 00 38 70 E1 1A 02 0A EE 00 00 07 A4 00 00 0B A3 A4 3E C2 C5 02 01 B1 E5 79 18 19 C5 02 04 EC EC 42 EE 92 C5 02 04 EB D9 5E 8D 54 C5 02 04 EB D9 69 B1 74

Decode Upload Clear Samples

216.27.185.42 → 192.168.50.50 NTP NTP Version 3, symmetric passive Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	D0	59	6C	40	4E	00	0C	41	B2	B2	53	00	00	45	00
00	5C	0A	22	40	00	32	11	22	5E	08	18	B9	3A	C0	A8
32	32	00	7B	00	7B	00	38	70	E1	1A	02	0A	EE	00	00
07	00	00	A3	A4	3E	C2	C5	02	01	B1	E5	79	18	19	C5
18	00	00	A3	A4	3E	C2	C5	02	01	B1	E5	79	18	19	C5
0D	00	00	A3	A4	3E	C2	C5	02	01	B1	E5	79	18	19	C5
0D	00	00	A3	A4	3E	C2	C5	02	01	B1	E5	79	18	19	C5

User Datagram Protocol
Src Port: 123
Dst Port: 123
Length: 8 bytes

4 protocols in packet (136 bytes):
Ethernet IPv4 UDP NTP

Filter tree... Width: 16 bytes Export Share

SET 2 Tracing a TCP Session

An FTP server response transported over TCP, peeled apart layer by layer.

Exercise 4 — Sequence Number

192.168.1.231 delivers the FTP welcome message to 192.168.1.182, with the sequence number called out.

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231] Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	3B	00	13	20	9F	04	2F	08	00	45	10
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	B6	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	00	00	00	00	00	00	00	00	00	00
47	F6	32	32	30	30	30	30	30	30	30	30	30	30	30	30
33	2E	30	61	20	20	20	20	20	20	20	20	20	20	20	20
46	54	50	44	20	20	20	20	20	20	20	20	20	20	20	20
65	72	76	65	72	72	72	72	72	72	72	72	72	72	72	72
31	2E	32	33	31	50	00	0A	00	00	00	00	00	00	00	00

Transmission Control Protocol
Sequence Number: 0x8364ffe
Field offset: [38-41]
Field length: 4
Byte offset: 38

4 protocols in packet (136 bytes):
Ethernet IPv4 TCP FTP

Exercise 4 (cont.) — Acknowledgment Number

The identical packet, this time with the acknowledgment number highlighted.

Wireshark packet capture showing the Acknowledgment Number field highlighted. The packet is an FTP response from 192.168.1.231 to 192.168.1.182. The Acknowledgment Number is 0x04846eda.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	3B	00	13	20	9F	04	2F	08	00	45	10
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	B6	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33	2E	30	61	20	53	65	72	76	65	72	20	28	50	72	6F
46	54	50	44	20	41	6E	6F	6E	79	60	6F	75	73	20	53
65	72	76	65	72	29	20	58	31	39	32	2E	31	36	38	2E
31	2E	32	33	31	5D	00	0A								

4 protocols in packet (136 bytes):

Ethernet IPv4 TCP FTP X

Transmission Control Protocol
Acknowledgment Number: 0x04846eda
Field offset: [42-45]
Field length: 4
Byte offset: 42

Exercise 5 — IPv4 Header

Moving up a layer on the same packet: the source and destination IP addresses.

Wireshark packet capture showing the IPv4 header fields highlighted. The packet is an FTP response from 192.168.1.231 to 192.168.1.182. The source IP is 192.168.1.231 and the destination IP is 192.168.1.182.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	3B	00	13	20	9F	04	2F	08	00	45	10
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	B6	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33	2E	30	61	20	53	65	72	76	65	72	20	28	50	72	6F
Internet Protocol Version 4															
Src: 192.168.1.231															
Dst: 192.168.1.182															
Length: 20 bytes															
		44	20	41	6E	6F	6E	79	60	6F	75	73	20	53	
		65	72	29	20	58	31	39	32	2E	31	36	38	2E	
		33	31	5D	00	0A									

4 protocols in packet (136 bytes):

Ethernet IPv4 TCP FTP X

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231]

+. Ask AI about this packet

Filter tree... Width: 16 bytes Export Share

SET 3 Comparing UDP and TCP

The same packets revisited, spotlighting different fields each round.

Exercise 6 — TCP Ports and Length

The FTP response again, this time showing source port, destination port, and segment length together.

05 A8 99 67 00 00 01 01 08 0A 3C F6 A9 AA 1E 8D 47 F6 32 32 30 20 50 72 6F 46 54 50 44 20 31 2E 33 2E 30 61 20 53 65 72 76 65 72 20 28 50 72 6F 46 54 50 44 20 41 6E 6F 6E 79 6D 6F 75 73 20 53 65 72 76 65 72 29 20 58 31 39 32 2E 31 36 38 2E 31 2E 32 33 31 5D 0D 0A

Decode Upload Clear Samples

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231] Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	3B	00	13	20	9F	04	2F	08	00	45	10
00	7A	EG	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	B6	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33					53	65	72	76	65	72	20	28	50	72	6F
46					41	6E	6F	6E	79	6D	6F	75	73	20	53
05					29	20	58	31	39	32	2E	31	36	38	2E
33					11	5D	0D	0A							

Transmission Control Protocol
Src Port: 21
Dst Port: 62014
Len: 70
Length: 32 bytes

4 protocols in packet (4 to bytes):
Ethernet IPv4 TCP FTP X

Exercise 7a — UDP Ports

Back to the DNS request from Exercise 1 — the UDP header expanded to show the port pairing (59198 → 53).

H Hex Packet Decoder 12,107,385 packets decoded Theme

52 54 00 7E 42 A5 52 54 00 A6 11 53 08 00 45 00 00 4A 4A 48 00 00 3F 11 F9 D7 C0 A8 38 0B C2 BE 7C 11 E7 3E 00 35 00 36 37 CB E4 A3 00 10 00 01 00 00 00 00 01 03 6E 73 32 0A 68 69 67 68 73 65 63 75 72 65 02 72 75 00 00 01 00 01 00 00 29 10 00 00 00 80 00 00 00

Decode Upload Clear Samples

192.168.56.11 → 194.190.124.17 DNS Standard query 0xe4a3 A ns2.highsecure.ru OPT Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
52	54	00	7E	42	A5	52	54	00	A6	11	53	08	00	45	00
00	4A	4A	48	00	00	3F	11	F9	D7	C0	A8	38	0B	C2	BE
7C	11	E7	3E	00	35	00	36	37	CB	E4	A3	00	10	00	01
00					01	03	6E	73	32	0A	68	69	67	68	73
65					02	72	75	00	00	01	00	01	00	00	29
10					00	00	00	00							

User Datagram Protocol
Src Port: 59198
Dst Port: 53
Length: 10 bytes

4 protocols in packet (4 to bytes):
Ethernet IPv4 UDP DNS X

Exercise 7b — TCP Ports

The FTP response's TCP header shown for side-by-side comparison.

05 A8 99 67 00 00 01 01 08 0A 3C F6 A9 AA 1E 8D 47 F6 32 32 30 20 50 72 6F 46 54 50 44 20 31 2E 33 2E 30 61 20 53 65 72 76 65 72 20 28 50 72 6F 46 54 50 44 20 41 6E 6F 6E 79 6D 6F 75 73 20 53 65 72 76 65 72 29 20 58 31 39 32 2E 31 36 38 2E 31 2E 32 33 31 50 00 0A

Decode Upload Clear Samples

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231] Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	38	00	13	20	9F	04	2F	00	00	45	18
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	86	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33	2E	30	61	20	53	65	72	76	65	72	20	28	50	72	6F
46	54	50	44	20	41	6E	6F	6E	79	6D	6F	75	73	20	53
65	72	76	65	72	29	20	58	31	39	32	2E	31	36	38	2E
31	2E	32	33	31	50	00	0A								

Transmission Control Protocol
Src Port: 21
Dst Port: 62014
Len: 70
Length: 32 bytes

4 protocols in packet (136 bytes):
Ethernet IPv4 TCP FTP X

Exercise 8a — Sequence Number (revisited)

The FTP segment once more, this time pairing the sequence number with its byte offset.

00 17 F2 2E 52 38 00 13 20 9F 04 2F 08 00 45 10 00 7A E6 5E 40 00 40 06 CF 21 C0 A8 01 E7 C0 A8 01 06 00 15 F2 3E 83 6F 4F FE 04 84 6E DA 80 18 05 A8 99 67 00 00 01 01 08 0A 3C F6 A9 AA 1E 8D 47 F6 32 32 30 20 50 72 6F 46 54 50 44 20 31 2E 33 2E 30 61 20 53 65 72 76 65 72 20 28 50 72 6F 46 54 50 44 20 41 6E 6F 6E 79 6D 6F 75 73 20 53 65 72 76 65 72 29 20 58 31 39 32 2E 31 36 38 2E 31 2E 32 33 31 50 00 0A

Decode Upload Clear Samples

192.168.1.231 → 192.168.1.182 FTP Response: 220 ProFTPD 1.3.0a Server (ProFTPD Anonymous Server) [192.168.1.231] Ask AI about this packet

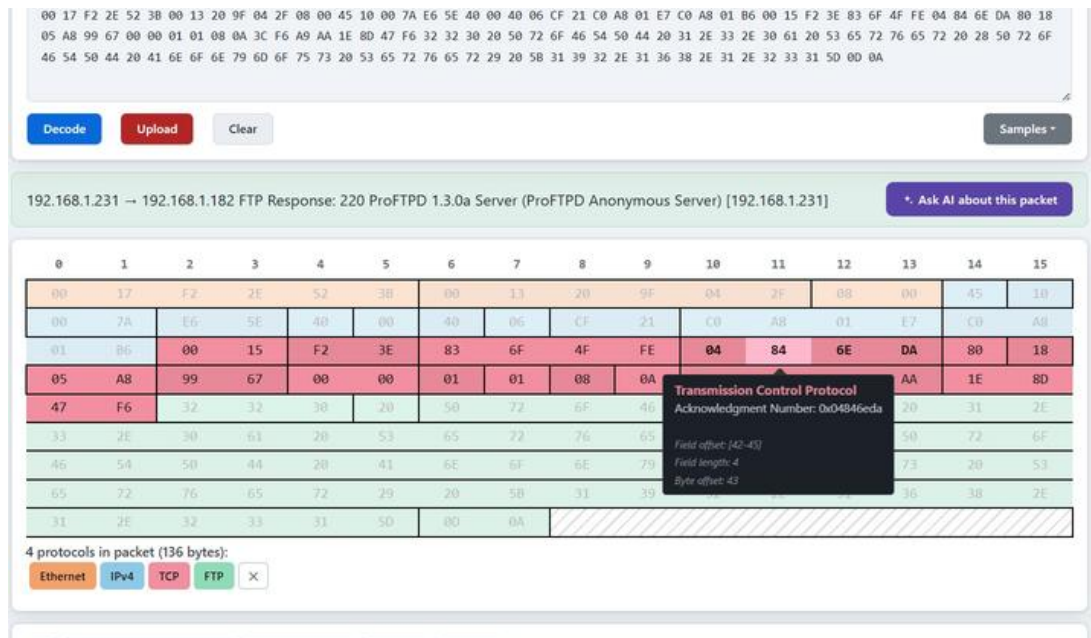
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	17	F2	2E	52	38	00	13	20	9F	04	2F	00	00	45	18
00	7A	E6	5E	40	00	40	06	CF	21	C0	A8	01	E7	C0	A8
01	86	00	15	F2	3E	83	6F	4F	FE	04	84	6E	DA	80	18
05	A8	99	67	00	00	01	01	08	0A	3C	F6	A9	AA	1E	8D
47	F6	32	32	30	20	50	72	6F	46	54	50	44	20	31	2E
33	2E	30	61	20	53	65	72	76	65	72	20	28	50	72	6F
46	54	50	44	20	41	6E	6F	6E	79	6D	6F	75	73	20	53
65	72	76	65	72	29	20	58	31	39	32	2E	31	36	38	2E
31	2E	32	33	31	50	00	0A								

Transmission Control Protocol
Sequence Number: 0x836f4ffe
Field offset: [38-41]
Field length: 4
Byte offset: 39

4 protocols in packet (136 bytes):
Ethernet IPv4 TCP FTP X

Exercise 8b — Acknowledgment Number (revisited)

Same segment, now showing the acknowledgment number alongside its byte offset.

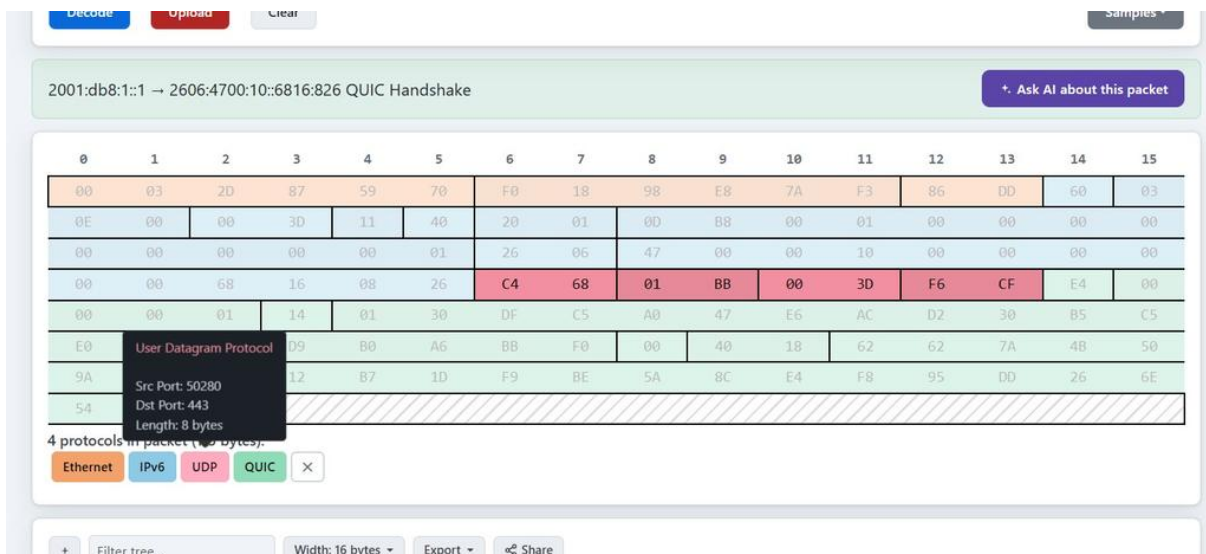


SET 4 Past UDP and TCP: Exploring QUIC

A QUIC connection handshake traveling over UDP on IPv6.

Exercise 10 — QUIC Handshake

Endpoints 2001:db8:1::1 and 2606:4700:10::6816:826 initiate a QUIC handshake, with the UDP header fields shown.



Exercise 10 (cont.) — Complete Frame

The entire 115-byte frame broken down across all layers: Ethernet, IPv6, UDP, and QUIC — checksum field highlighted.

DecodeUploadClear

Samples ▾

2001:db8:1::1 → 2606:4700:10::6816:826 QUIC Handshake

Ask AI about this packet

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
00	03	2D	87	59	70	F0	18	98	E8	7A				60	03
0E	00	00	3D	11	40	20	01	0D	B8	00				00	00
00	00	00	00	00	01	26	06	47	00	00				00	00
00	00	68	16	08	26	C4	68	01	BB	00	3D	F6	CF	E4	00
00	00	01	14	01	30	DF	C5	A0	47	E6	AC	D2	30	B5	C5
E0	47	CE	D9	B0	A6	BB	F0	00	40	18	62	62	7A	4B	50
9A	40	68	12	B7	1D	F9	BE	5A	8C	E4	F8	95	DD	26	6E
54	56	7D													

4 protocols in packet (115 bytes):

EthernetIPv6UDPQUIC

+Filter tree...

Width: 16 bytes ▾

Export ▾

Share

Frame 1: 115 bytes on wire (920 bits)